Exercise Sheet 15

Exercises 15.1–15.5 are the core work. Exercise 15.6 is an optional extension. We use the conventions for $L^2(\mathbb{R})$ from Lecture 15. A piecewise continuous function f with finitely many jumps in each bounded interval and finite $\int_{\mathbb{R}} |f|^2\,dx$ defines a vector of $L^2(\mathbb{R})$. Inner products are integrals. Changing finitely many values does not change the vector. The translation representation is $(\lambda(t)f)(x) = f(x-t)$, and A denotes its generator.

The formulas below for reflection, modulation and dilation are first defined on $C_c(\mathbb{R})$. After norm preservation is proved, use their continuous extensions to $L^2(\mathbb{R})$. Approximation by $C_c(\mathbb{R})$ gives the same formulas on the piecewise continuous vectors described above.

Exercise 15.1. For $t > 0$ and $N \geq 1$ let $I_k = [kt, (k+1)t)$ and

$$f_N = \sum_{k=0}^{N-1} (-1)^k 1_{I_k} \in L^2(\mathbb{R}).$$

(a) Compute $\|f_N\|^2$ and $\|\lambda(t)f_N - f_N\|^2$.
(b) Deduce that $\|\lambda(t) - I\| = 2$ for every $t \neq 0$. Is λ norm continuous?
(c) Let $B = -d/dx$ with domain $\mathcal{S}(\mathbb{R})$. For $a > 0$ let $g_a(x) = e^{-ax^2/2}$. Show that $\|Bg_a\|^2 = \frac{a}{2}\|g_a\|^2$, and deduce that B is not bounded.
(d) Show that no bounded operator T on $L^2(\mathbb{R})$ satisfies $\lambda(t) = \sum_{n\geq 0} t^n T^n/n!$ for all $t \in \mathbb{R}$.

Exercise 15.2. Let $g(x) = \max(0, 1-|x|)$ and let $k = 1_{(-1,0)} - 1_{(0,1)}$, the derivative of g away from $-1, 0, 1$.

(a) For $0 < t \leq 1$ compute $\|\lambda(t)g - g\|^2$.
(b) Let $q_t = \frac{\lambda(t)g - g}{t} + k$. Show $\|q_t\|^2 = 2|t|$ for $0 < |t| \leq 1$. Conclude that $g \in D(A)$ with $Ag = -k$.
(c) For $0 < t \leq 1$ compute $\|\lambda(t)k - k\|^2$ and deduce $k \notin D(A)$.
(d) Deduce that $g \in D(A)$, $g \notin L^2(\mathbb{R})^\infty$ and $g \notin \mathcal{S}(\mathbb{R})$.

Exercise 15.3. Let $\mathcal{H}$ be a Hilbert space with orthonormal basis $(e_n)_{n\in\mathbb{Z}}$. For example, take L^2 of the circle $\mathbb{R}/2\pi\mathbb{Z}$ with normalised Haar measure $d\theta/(2\pi)$ and $e_n(\theta) = e^{in\theta}$. Only the orthonormal basis is used below. Write $v = \sum_n c_n e_n$ with $c_n = \langle e_n, v\rangle$, and define

$$U(t)v = \sum_{n\in\mathbb{Z}} e^{-int} c_n e_n.$$

On the circle this is rotation, $(U(t)f)(\theta) = f(\theta - t)$. Let A be the generator of U.

(a) Show that U is a unitary representation of $\mathbb{R}$.
(b) Show that $D(A) = \{v : \sum_n n^2|c_n|^2 < \infty\}$ and $Av = \sum_n (-in)c_n e_n$.
(c) Show directly that $A^* = -A$.
(d) Let $\mathcal{F}$ be the span of the e_n. Show directly that the closure of $A|_{\mathcal{F}}$ is A.

Exercise 15.4. Let U be a strongly continuous unitary representation of $\mathbb{R}$ on $\mathcal{H}$ with generator A. You may use these facts. For continuous compactly supported $F\colon \mathbb{R} \to \mathcal{H}$, the integral $\int_{\mathbb{R}} F(t)\,dt$ exists as a norm limit of Riemann sums. It is linear and satisfies

$$T\int F = \int TF, \qquad \left\|\int F\right\| \le \int \|F\|, \qquad \int F(t+s)\,dt = \int F(t)\,dt$$

for bounded T and $s \in \mathbb{R}$. For $\varphi \in C_c^\infty(\mathbb{R})$ and $v \in \mathcal{H}$ put

$$U(\varphi)v = \int_{\mathbb{R}} \varphi(t)\,U(t)v\,dt.$$

(a) Show $U(s)U(\varphi)v = U(\varphi_s)v$, where $\varphi_s(t) = \varphi(t-s)$.

(b) Show $U(\varphi)v \in D(A)$ and $AU(\varphi)v = -U(\varphi')v$.

(c) Show $U(\varphi)v \in \mathcal{H}^\infty$.

(d) Let $\varphi \ge 0$ have support in $[-1,1]$ and $\int \varphi = 1$, and put $\varphi_n(t) = n\varphi(nt)$. Show $U(\varphi_n)v \to v$. Conclude that $\mathcal{H}^\infty$ and $D(A)$ are dense.

Exercise 15.5. On $L^2(\mathbb{R})$ define modulations $(M(s)f)(x) = e^{isx}f(x)$ for $s \in \mathbb{R}$. On $\mathcal{S}(\mathbb{R})$ let $B = -d/dx$ and $(Cf)(x) = ix\,f(x)$.

(a) Show that M is a unitary representation of $\mathbb{R}$.

(b) Show that $\mathcal{S}(\mathbb{R})$ consists of smooth vectors for M and that $dM(1) = C$ on $\mathcal{S}(\mathbb{R})$.

(c) Compute $[B,C]f = BCf - CBf$ for $f \in \mathcal{S}(\mathbb{R})$.

(d) Prove the Weyl relation $\lambda(t)M(s) = e^{-ist}M(s)\lambda(t)$. Differentiate it in s at $s = 0$ on $f \in \mathcal{S}(\mathbb{R})$, and check the result directly.

Exercise 15.6 (Optional extension). On $L^2(\mathbb{R})$ define dilations $(\delta(s)f)(x) = e^{-s/2}f(e^{-s}x)$ for $s \in \mathbb{R}$. Let B and C be as in Exercise 15.5.

(a) Show that δ is a unitary representation of $\mathbb{R}$.

(b) For $f \in \mathcal{S}(\mathbb{R})$ compute the pointwise derivative $Ef = \frac{d}{ds}\big|_{s=0}\delta(s)f$. Show that E is skew-symmetric on $\mathcal{S}(\mathbb{R})$.

(c) Show $E = \frac{1}{2i}(BC + CB)$ on $\mathcal{S}(\mathbb{R})$.